

Question 1 Sol.

$$|\psi\rangle = \sqrt{0.7}|0\rangle + (0.5 + 0.5i)|1\rangle$$

- (1) $|\psi\rangle$ is a valid normalized quantum state $\Leftrightarrow |\alpha|^2 + |\beta|^2 = 1$
 $|\sqrt{0.7}|^2 + |(0.5 + 0.5i)|^2 = 0.7 + 0.25 + 0.25 = 1.2 \neq 1$
 $\therefore |\psi\rangle$ is not a valid normalized quantum state

(2) $N^2 = 1.2 \Rightarrow N = \sqrt{1.2}$ Normalization Factor

$$|\psi_{\text{norm}}\rangle = \frac{\sqrt{0.7}}{\sqrt{1.2}}|0\rangle + \left(\frac{0.5}{\sqrt{1.2}} + \frac{0.5}{\sqrt{1.2}}i\right)|1\rangle$$

(3) Expectation value of Pauli-Z

$$\begin{aligned}\langle Z \rangle &= \langle \psi_{\text{norm}} | Z | \psi_{\text{norm}} \rangle = |\alpha|^2 - |\beta|^2 \\ &= \frac{0.7}{1.2} - \left(\frac{0.25}{1.2} + \frac{0.25}{1.2} \right) = \frac{0.7 - 0.5}{1.2} = \frac{2}{12} = \frac{1}{6}\end{aligned}$$

Question 2 Sol.

$$|\psi_{\text{norm}}\rangle = \sqrt{\frac{7}{12}}|0\rangle + \sqrt{\frac{5}{12}}e^{i\pi/4}|1\rangle$$

(1) $|\psi\rangle = \cos\left(\frac{\theta}{2}\right) + e^{i\phi}\sin\left(\frac{\theta}{2}\right)$

$$\Rightarrow \cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{7}{12}}$$

$$\theta = 2 \arccos\left(\sqrt{\frac{7}{12}}\right) \approx 1.4 \text{ rad} \quad \text{Polar Angle}$$

$$\Rightarrow e^{i\phi} \sin\left(\frac{\theta}{2}\right) = e^{i\frac{\pi}{4}} \cdot \sqrt{\frac{5}{12}}$$

$$\phi = \frac{\pi}{4}$$

Azimuthal Angle

(2) Bloch Vector $\vec{r} = (x, y, z)$

From Q.1 (3) $z = \frac{1}{6}$

$$|\psi\rangle = \frac{\sqrt{0.7}}{\sqrt{1.2}}|0\rangle + \left(\frac{0.5}{\sqrt{1.2}} + \frac{0.5}{\sqrt{1.2}}i\right)|1\rangle$$

$$\alpha = \frac{\sqrt{0.7}}{\sqrt{1.2}} \Rightarrow \text{Complex Conjugate } \alpha^* = \frac{\sqrt{0.7}}{\sqrt{1.2}}$$

$$\begin{aligned} 2\alpha^*\beta &= 2 \left(\left(\frac{\sqrt{0.7}}{\sqrt{1.2}} \right) \left(\frac{0.5}{\sqrt{1.2}} + \frac{0.5}{\sqrt{1.2}}i \right) \right) \\ &= 2 \left(\frac{\sqrt{0.7} \cdot 0.5}{1.2} + \frac{\sqrt{0.7} \cdot 0.5}{1.2}i \right) \\ &= \frac{\sqrt{0.7}}{1.2} + \frac{\sqrt{0.7}}{1.2}i \end{aligned}$$

$$x = \text{Re}(2\alpha^*\beta) = \frac{\sqrt{0.7}}{1.2}$$

$$y = \text{Im}(2\alpha^*\beta) = \frac{\sqrt{0.7}}{1.2}$$

$$\therefore \vec{r} = \left(\frac{\sqrt{0.7}}{1.2}, \frac{\sqrt{0.7}}{1.2}, \frac{1}{6} \right)$$

(3) Solution in code solution folder.